Physics World Models for Computational Imaging: A Universal Physics-Information Law for Recoverability, Carrier Noise, and Operator Mismatch

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Correspondence: [your-email] | Date: February 14, 2026

Abstract

Computational imaging couples physical carriers (photons, electrons, spins, acoustic waves, particles) with algorithmic reconstruction. Yet failures are frequently misattributed to the reconstruction algorithm, when the true bottleneck lies in physics: insufficient measurement diversity, inadequate carrier information budget under noise constraints, or hidden mismatch between the assumed and true forward operator. We introduce PWM (Physics World Model), a universal physics + reconstruction + diagnosis instrument that compiles diverse imaging modalities into a canonical OperatorGraph and executes deterministic multi-agent diagnosis and bounded operator correction. PWM formalizes a universal physics-information law that decomposes reconstruction outcomes into three measurable gates: (i) Recoverability-operator geometry under a declared prior class and measurement budget; (ii) Carrier information budget-likelihood and noise channel grounded in the physical carrier and sensor pipeline; and (iii) Operator mismatch-bounded uncertainty and identifiability of forward parameters. PWM produces a reproducible RunBundle containing measurements, reconstructions, calibration trajectories, and structured diagnoses, enabling modality-agnostic failure attribution and intervention. We propose evaluation protocols for zero-shot physical diagnosis on unseen modalities from OperatorGraph specification and physics tier alone, and for mapping impossibility boundaries where physics makes recovery mathematically unattainable. PWM reframes computational imaging from incremental PSNR improvements into a principled science of physical generalization.

Keywords: computational imaging; physics world model; operator graph; forward-model mismatch; calibration; compressed sensing; noise modeling; physical generalization; reproducibility

1. Introduction

Computational imaging is built on an intimate coupling between physics and computation: an acquisition system encodes a scene or object into measurements through a forward physical process, and a computational pipeline reconstructs the underlying signal. Across microscopy, compressive imaging, medical imaging, coherent phase retrieval, diffuse optical tomography, ultrasound/photoacoustics, and particle-based modalities, progress is often reported as improvements in reconstruction metrics such as PSNR or SSIM.

However, practical failures rarely stem from the reconstruction algorithm alone. In real systems, three physical bottlenecks dominate outcomes:

(A) Recoverability: the acquisition design may not contain sufficient independent information, under a declared prior class, to permit stable recovery.

(B) Carrier information budget: the physical carrier and sensor chain impose noise, dose, saturation, and bandwidth constraints that set an achievable performance ceiling.

(C) Operator mismatch: hidden departures between the assumed and true forward operator-misalignment, PSF drift, dispersion errors, coil sensitivities, speed-of-sound variations-can induce structured bias that overwhelms improvements in the recon solver.

Because these bottlenecks are entangled, debugging is often ad hoc: researchers swap solvers, tweak priors, or overfit to a specific dataset. What is missing is a universal theory and instrument for physical generalization: a framework that (i) predicts whether a modality can ever reconstruct well under stated constraints, (ii) attributes failure to the correct physical gate, and (iii) recommends corrective actions at the level of acquisition, noise handling, or forward-model calibration.

In this work, we introduce PWM (Physics World Model), an open and reproducible physics + reconstruction + diagnosis toolkit for computational imaging. PWM turns a prompt/specification or measured data with an imperfect operator into a fully reproducible run: simulate or load measurements, optionally fit bounded operator parameters, reconstruct, diagnose, recommend actions, and export evidence as a RunBundle with a viewer. PWM is designed to be deterministic by default and extensible through plugins and modality adapters.

2. PWM as a Universal Instrument

PWM is built around three commitments:

(1) A universal intermediate representation (IR) for forward models.

(2) A physics fidelity ladder that enables controlled approximation and physical generalization.

(3) A deterministic diagnosis contract that separates algorithmic failure from physical impossibility.

2.1 OperatorGraph IR

PWM compiles each imaging modality into an OperatorGraph: a directed graph of physics primitives that implement a forward operator (and, when applicable, an adjoint or differentiation interface). Each node encodes its tensor signatures and tags. This IR is the anchor that makes modality-agnostic reasoning possible: it provides a standardized interface for simulation, calibration, reconstruction, and diagnosis.

2.2 Physics Fidelity Ladder

PWM uses a tiered physics ladder applied per node (not globally). Tier 0 captures geometric/ray regimes; Tier 1 uses wave/field approximations (e.g., paraxial or Fourier optics, linearized transport); Tier 2 invokes full-physics transport/scattering (e.g., Maxwell, full wave equation, Monte Carlo, quantum corrections); Tier 3 uses learned surrogates that emulate Tier 2 and must provide calibrated uncertainty. A TierPolicy selects the cheapest tier meeting the accuracy and budget requirements, and higher tiers can be used to validate or refine calibration.

2.3 Multi-agent execution and reproducibility

PWM orchestrates a portfolio of deterministic agents (with optional LLM narration constrained to registry IDs). Key agents include a RecoverabilityAgent, a carrier-aware noise-modeling agent, a MismatchAgent, an AnalysisAgent, and PreFlight/continuity checking. Each run exports a RunBundle capturing measurements, reconstructions, metrics, internal diagnostic state, and agent snapshots, enabling complete reproducibility and auditability.

3. A Universal Physics-Information Law

We model a computational imaging experiment as a carrier-constrained information channel:

y = F_{theta*}(x) + n,

where x is the latent signal, F_{theta*} is the true forward operator parameterized by physical/system parameters theta*, and n is noise induced by the physical carrier and sensor pipeline. PWM formalizes the following law:

Universal Physics-Information Law (Triad Decomposition). For a declared signal class (prior) and an acquisition budget, reconstruction outcomes are governed by three measurable gates:

Gate A: Recoverability (Compression/Geometry Gate). Whether the operator under the declared prior admits a stable inverse at the measurement budget and coding diversity. This is a property of operator geometry and signal class, independent of a specific recon algorithm.

Gate B: Carrier Information Budget (Source/Noise Gate). Whether the physical carrier and detector pipeline provide sufficient information after accounting for noise, dose, saturation, and noise folding under compression. This gate provides an achievable ceiling even for optimal estimators.

Gate C: Operator Mismatch (Hidden Physics Gate). Whether differences between assumed and true operators are small enough to ignore or, if not, whether the mismatched parameters are identifiable and correctable under the acquisition. Mismatch induces structured bias rather than i.i.d. noise.

PWM operationalizes this law by enforcing a standard diagnosis contract (TriadReport) that scores and explains each gate and proposes minimal physical interventions.

3.1 TriadReport: a standardized diagnosis contract

Gate	Question	Measurable outputs	Typical interventions
A: Recoverability	Can this system reconstruct under stated prior and budget?	phase diagram; conditioning/coherence proxies; PASS/WARN/FAIL	increase measurements; redesign codes; add diversity; change task/prior
B: Carrier budget	Is post-compression SNR/likelihood sufficient?	noise family + parameters; SNR-after-compression; ceiling estimate	increase dose/exposure; reduce compression; change sensor/readout; denoise model
C: Mismatch	Is operator error identifiable/correctable?	mismatch suspicion score; theta_hat ± uncertainty; identifiability flag	bounded calibration; add calibration signals; increase diversity; use higher tier validation

4. Zero-shot Physical Diagnosis and Generalization

To move beyond modality-specific tuning, we define a rigorous zero-shot protocol.

Definition (Zero-shot physical diagnosis). A modality family is held out during development of modality registries and calibration tables. At test time, PWM receives only (i) an OperatorGraphSpec encoding the forward model family and tensor signatures, (ii) physics tier tags and TierPolicy constraints, (iii) carrier type and sensor metadata, (iv) declared prior class and measurement budget. PWM must output (a) which gate (recoverability, carrier budget, mismatch) is the dominant bottleneck and (b) a minimal intervention. Reconstruction is performed by a generic solver portfolio; the novelty is that PWM selects, constrains, and audits the pipeline using physics.

Metrics. We evaluate: (1) bottleneck classification accuracy; (2) intervention effectiveness measured as delta in task metrics after applying PWM's recommended intervention; (3) calibration confidence and identifiability, quantified by uncertainty intervals and sensitivity analysis.

This protocol turns 'physical reasoning' into a falsifiable claim: PWM must generalize across modalities through its physics IR and triad law, not through ad hoc solver tuning.

5. Quantifiable Physical Limits and Impossibility Boundaries

High-impact scientific claims require quantifiable limits. PWM maps impossibility boundaries in a three-axis space:

axis 1: measurement budget and diversity (recoverability);

axis 2: carrier information budget (noise/dose/exposure);

axis 3: mismatch magnitude and identifiability.

For each modality and declared prior class, PWM sweeps these axes to identify surfaces where stable recovery collapses across solver families. These boundary surfaces are reported as phase diagrams and sensitivity curves, and they enable two practical outcomes:

(i) predicting when reconstruction is mathematically unattainable (no-go regions), and

(ii) prescribing which physical knob (more measurements, more dose, or calibration) most efficiently moves the system into a feasible region.

Unlike PSNR-focused evaluation, boundary mapping produces a general theory of physical generalization and failure attribution that transfers across imaging domains.

6. Results: a minimal but decisive evaluation suite

This manuscript is designed to be filled with results from a compact set of experiments that jointly verify universality, zero-shot diagnosis, and real-world impact. Below we specify the evaluation suite required for high-impact review.

6.1 Breadth benchmark (synthetic + standardized)

PWM provides prompt-driven workflows for 64 validated modalities and a benchmark runner. We report: (i) baseline reconstruction across solver portfolios, (ii) TriadReport summaries, and (iii) the distribution of dominant bottlenecks across modality categories.

6.2 Operator correction across modalities (mismatch gate)

PWM supports operator correction in 16 modalities with verified improvements (>0.5 dB). We report: (i) calibration trajectories, (ii) theta estimates with uncertainty, and (iii) before/after reconstructions with residual diagnostics.

6.3 Zero-shot diagnosis on held-out modality families

We hold out at least two modality families (e.g., coherent phase retrieval / ptychography and photoacoustics) and evaluate whether PWM correctly attributes failures and recommends interventions that improve outcomes.

6.4 High-stakes real-world demonstrations

We recommend two demonstrations where diagnosis and correction change what experiments are feasible: (i) biological microscopy in a low-dose regime (e.g., SIM-like), and (ii) coherent imaging for sub-pixel defect detection (ptychography/semiconductor metrology). Performance should be reported in task metrics (detection precision/recall, resolution characterization, reproducibility), not only PSNR.

Figure plan (main figures):

Figure 1: PWM overview pipeline and RunBundle reproducibility contract.

Figure 2: OperatorGraph IR and physics fidelity ladder with tier selection per node.

Figure 3: Triad law boundary surfaces and failure attribution across modalities.

Figure 4: Real data case studies showing PWM’s diagnosis and intervention.

7. Discussion

PWM reframes computational imaging as a science of physical generalization. The triad law clarifies why 'better networks' often fail to translate across instruments: recoverability, carrier budgets, and mismatch create hard limits and structured biases. The OperatorGraph + physics tier approach provides a universal interface to perform diagnosis and intervention across carriers and modalities.

Limitations and future work include: richer uncertainty quantification for Tier-3 learned surrogates; broader coverage of nonlinear and stochastic operators; and standardized acquisition metadata schemas for measured datasets. Nonetheless, the combination of a universal IR, a tiered physics ladder, a deterministic triad diagnosis contract, and operator correction mode positions PWM as a reusable scientific instrument, not merely a reconstruction method.

8. Methods (summary)

8.1 OperatorGraph specification

Each modality is defined by an OperatorGraphSpec comprising nodes (physics primitives), edges (tensor specs), and parameter specs with bounds. Nodes expose forward evaluation and, when available, adjoint and/or differentiation interfaces.

8.2 TierPolicy and validation

Each node is tagged with a physics tier. The TierPolicy selects an execution plan under compute and accuracy constraints, and higher tiers can be used to validate low-tier approximations during calibration.

8.3 Agent contracts and determinism

Agents produce structured outputs and must run deterministically without LLMs. LLM narration is optional and restricted to registry identifiers.

8.4 Operator correction mode

For measured data with imperfect operators, PWM estimates bounded forward parameters theta using a calibration loop (coarse search + refinement), reconstructs with the corrected operator, and exports calibration evidence.

8.5 Metrics

Report metrics must include task-relevant measures where applicable (detection, resolution, reproducibility), alongside standard reconstruction metrics.

Data availability

All data required to reproduce the main conclusions (the 'minimum dataset') will be made available in a public repository upon publication, including synthetic benchmarks, measurement data used in real-world case studies, and derived artifacts necessary to verify the reported results. Access conditions and any restrictions (e.g., industrial data) will be transparently described in this section.

Code availability

PWM code and scripts required to reproduce the figures will be made available in a public repository upon publication. The repository will include versioned modality registries, solver registries, and scripts to generate RunBundles and reproduce the evaluation protocol.

Acknowledgements

[To be completed: funding sources, collaborators, and institutional support.]

Author contributions

Conceptualization: C.Y. Methodology: C.Y. Software: C.Y. Validation: C.Y. Writing - original draft: C.Y. Writing - review & editing: [to be completed].

Competing interests

The author declares no competing interests. [Update if needed.]

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10. Add modality-specific foundational references (SIM, ptychography, CASSI/SCI, MRI, CT, photoacoustic) based on the final evaluation suite.